INCH-POUND

MIL-DTL-23806/2D
w/AMENDMENT 1
13 January 2022
SUPERSEDING
MIL-DTL-23806/2D
22 February 2007

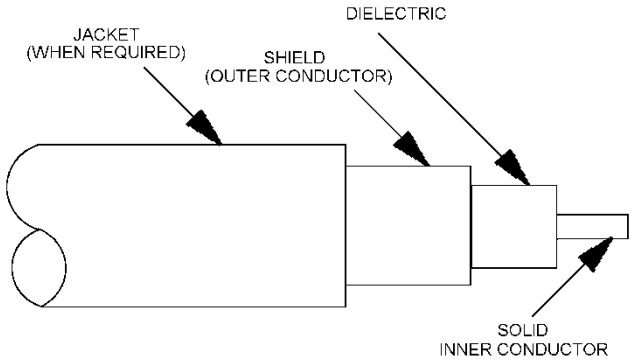
DETAIL SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, COAXIAL, SEMIRIGID, FOAM DIELECTRIC, .875 INCH, 50 AND 75 OHM, (RG-332/U, RG-333/U, RG-336/U and RG-306A/U)

Inactive for new design after 10 July 2000

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-23806.



Inches	mm
.001	0.03
.002	0.05
.004	0.10
.006	0.15
.058	1.47
.173	4.39
.288	7.32
.801	20.35
.875	22.23
1.052	26.72
9.0	229.0

Characteristics	50 ohms		75 ohms	
	Type I RG-332/U	Type II RG-333/U	Type I RG-336/U	Type II RG-306A/U
	Construction details			
Inner conductor Outside diameter	0.288 ± .003	0.288 ± .003	0.173 ± .002	0.173 ± .002
Outer conductor				
Inside diameter	0.801 ± .004	0.801 ± .004	0.801 ± .004	0.801 ± .004
Outside diameter	0.875 ± .006	0.875 ± .006	0.875 ± .006	0.875 ± .006
Jacket				
Minimum wall thickness		.058		.058
Maximum outside diameter		1.052		1.052
Minimum bending radius	9	9	9	9
Weight (pounds per 1000 feet, maximum)	480	545	306	376

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for information only.

FIGURE 1. Construction.

AMSC N/A FSC 6145

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:57Z
Check the source to verify that this is the current version before use.

ENGINEERING INFORMATION:

Continuous working voltage:

RG-332/U RG-333/U	RG-336/U RG-306A/U
3,080 V rms, max.	3,470 V rms, max.

Operating frequency range: 6 GHz, maximum.
Power rating: See figure 2.
Temperature rating: -55°C to +85°C.

REQUIREMENTS (RG-332/U and RG-333/U):

Design and construction: See figure 1.
Impedance: 50 $\pm$ 1.
Capacitance: 25 pF/ft, nominal.
Velocity of propagation: 81 $\pm$ 3 percent.
Dielectric strength (at 60 Hz): 6,200 V rms, minimum.
Dielectric constant: 1.5, nominal.
Attenuation: See figure 2.

REQUIREMENTS (RG-336/U and RG-306A/U):

Design and construction: See figure 1.
Impedance: 75 $\pm$ 2.
Capacitance: 16.5 pF/ft, nominal.
Velocity of propagation: 81 $\pm$ 3 percent.
Dielectric strength (at 60 Hz): 5,500 V rms, minimum.
Dielectric constant: 1.5, nominal.
Attenuation: See figure 2.

REQUIREMENTS (All cables):

The attenuation measurement for all cables is tested at 30 MHz, 400 MHz and 3 GHz; maximum attenuation is not to exceed 0.30 dB, 1.2 dB and 5.0 dB per 100 feet.

Voltage standing wave ratio (VSWR): See table I.

TABLE I. Voltage standing wave ratio.

Frequency range (MHz)	Initial (maximum)	After temperature cycling (maximum)	After cooling (maximum)
500 to 2000 2000 to 5000	1.15 1.20	1.18 1.25	1.18 1.25

Temperature cycling and bending test: Coiling mandrel, 27 inches in diameter, maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:57Z
Check the source to verify that this is the current version before use.

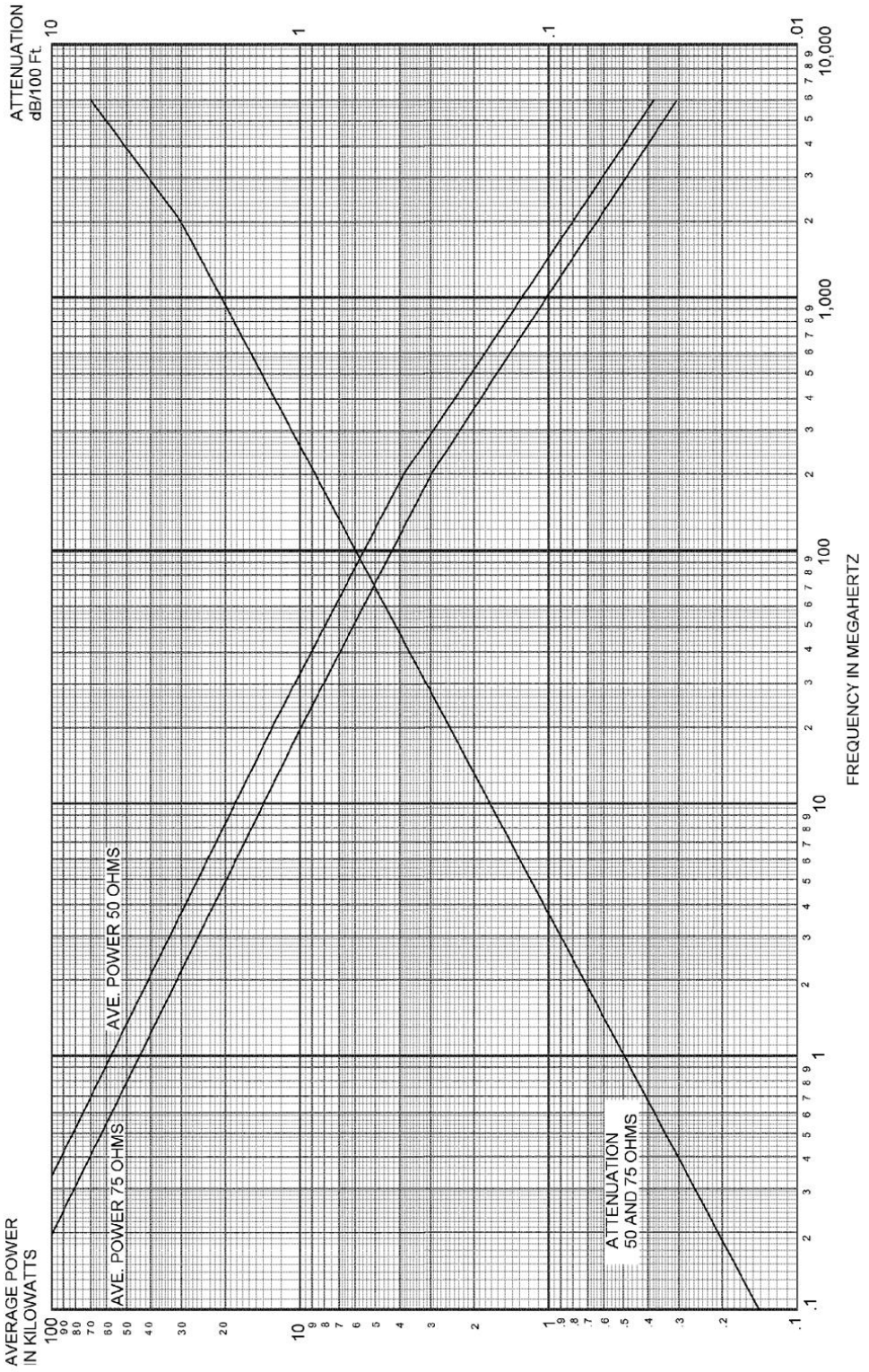


FIGURE 2. .875 inch foam dielectric coaxial cable (50 and 75 ohm).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:57Z
Check the source to verify that this is the current version before use.

Amendment Notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Reference documents. This document references MIL-DTL-23806.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

(Project 6145-2022-003)

Review activities:
Army – AR, AV, MI
Navy – MC, MS, SA, YD
Air Force – 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:57Z
Check the source to verify that this is the current version before use.